Low-dimensional gauge gaps: matrix confinement, conditional floors and scales

With supplied physical $\hbar > 0$, $m > 0$ and $g > 0$, the scalar $H = -\hbar^2\Delta/(2m) + g^2x^2y^2/2$ and the bosonic $SU(2)$ model with at least two matrices have compact resolvent and a simple positive ground state. Their excitation gap is a positive finite number times $E_* = \hbar^{4/3}g^{2/3}m^{-2/3}$. The proof uses transverse oscillator bounds, compactness and positivity improvement. The gauge-invariant sector inherits a positive gap; its first excitation need not equal the unrestricted one. These are quantum-mechanical spectral theorems, not construction of a continuum Yang–Mills theory.

A supplied **classical** transverse orbit-area floor gives a different result: it bounds energy sublevel sets and makes their phase-space volume finite. It does not define a quantum Hamiltonian, discretize levels, or by itself give a positive energy gap. The precise volume asymptotics, semiclassical estimates and dimensional comparisons below retain their conditional status. In four spacetime dimensions $1/g_{\mathrm{cl}}^2$ is already an action unit; its cancellation from the classical field equations does not mean that the action has no normalization. An action unit supplies neither an energy scale nor a gap there.

This treatment incorporates C131–C133 and the former G07/G08 comparison. Borrowed spectral theorems, elementary proofs, conditional classical restrictions, semiclassical estimates and open questions are labelled separately. Repository reviews are internal written checks; no independent peer review or literature novelty is claimed.

1. Observables, units and conventions

A gap is an energy difference of a specified self-adjoint operator. For an isolated simple ground state it is $E_1 - E_0$; a field-theoretic mass gap also requires the physical state space, vacuum, continuum construction and limiting order. A classical action floor instead takes an infimum of a specified action observable over an admitted class. Neither is a statement about all possible classical completions or instruments.

The dimensional and similarity proofs apply to any observable dimension d_X (C131): if the floor depends only on the fixed constants and d_X is outside their dimension span, a finite positive universal floor is impossible. Otherwise it is a dimensional monomial times a function of dimensionless ratios. An admitted similarity scaling one positive value arbitrarily close to zero gives zero infimum. These are necessary unit and class tests, not sufficient positivity arguments.

In the mechanical model g denotes its quartic coupling. For connection-valued fields use g_{cl} with $[1/g_{\mathrm{cl}}^2] = \text{action}\,\text{length}^{4-d}$; the dimensionless four-dimensional coupling is $g_4^2 = \hbar g_{\mathrm{cl}}^2$. The torus formula in §3.4 retains g for g_4 as explicitly stated there. Energy units are denoted E_*; they are unrelated to the dimensionless regulator $\epsilon = \hbar/S_0$, time mesh or record resolution. Write $h_P = 2\pi\hbar$ for Planck's constant.

2. The exact $\hbar$-scaling of spectra for homogeneous potentials

Theorem 3. Take $m, \lambda, \hbar > 0$. Let $V \geq 0$ be continuous on $\mathbb{R}^n$ and positively homogeneous of degree $k > -2$, $V(sx) = s^kV(x)$, and let $H = -\frac{\hbar^2}{2m}\Delta + \lambda V$ be the Friedrichs operator of its form on C_0^∞. With the unitary dilation $(U_\ell\psi)(x) = \ell^{-n/2}\psi(x/\ell)$ and

$$\ell^{k+2} = \frac{\hbar^2}{m\lambda}, \qquad E_* = \frac{\hbar^2}{m\ell^2} = \hbar^{\frac{2k}{k+2}}\,\lambda^{\frac{2}{k+2}}\,m^{-\frac{k}{k+2}},$$

one has $U_\ell^{-1}HU_\ell = E_*\,(-\frac{1}{2}\Delta + V)$. Every spectral datum of H, in particular any gap, is E_* times the corresponding pure number of $-\frac{1}{2}\Delta + V$.

Proof. $\Delta(U_\ell\psi) = \ell^{-2}U_\ell(\Delta\psi)$ and $V\,U_\ell\psi = \ell^kU_\ell(V\psi)$ on C_0^∞, which U_ℓ preserves; so $U_\ell^{-1}HU_\ell = \frac{\hbar^2}{m\ell^2}(-\frac{1}{2}\Delta) + \lambda\ell^kV$, and the choice of ℓ equates the two coefficients. The identity

passes to the Friedrichs extensions because U_ℓ maps the form on the core to the transformed form. Unitary equivalence preserves spectra and multiplicities. □

Proposition 4 (classical similarity). For the classical Hamiltonian $|p|^2/2m + \lambda V(x)$ with V homogeneous of degree k, the map $x \mapsto sx$, $t \mapsto s^{1-k/2}t$ carries solutions to solutions, multiplies energies by s^k and the action along a trajectory by $s^{1+k/2}$ (Landau–Lifshitz, *Mechanics* §10; verified against the equation of motion: $m\ddot{x}$ scales as $s/s^{2-k} = s^{k-1}$ and $\lambda\nabla V$ as s^{k-1}). Hence for $k > 0$ the range of the classical Hamiltonian is $[0, \infty)$: the classical energy floor is zero, and the classical action floor is zero by the similarity theorem cited in §1.

For $k > 0$, any *proved* gap scales to zero as $\hbar \to 0$ or $\lambda \to 0$; the dilation alone does not prove that a gap exists. The oscillator ($k = 2$), quartic model ($k = 4$) and a box of supplied length give units $\hbar\sqrt{\lambda/m}$, $\hbar^{4/3}\lambda^{1/3}m^{-2/3}$ and $\hbar^2/(m\ell^2)$ respectively. The action-critical degree $k = -2$ is excluded from the theorem and treated in §6. Its classical action is scale invariant, while $m\lambda$ has action-squared units.

For the separately defined attractive Coulomb problem, the familiar energy unit is $m\lambda^2/\hbar^2$; this singular negative potential does not satisfy the theorem's $V \geq 0$ continuity assumptions. Fixed-index hydrogenic energies diverge negatively as $\hbar \to 0$. Their accumulation at zero is the *large principal-index* limit at fixed $\hbar$, not the same limit.

3. Quantum confinement of bosonic matrix valleys

3.1 The scalar model

Theorem 5. Let $H = -\frac{\hbar^2}{2m}\Delta + \frac{g^2}{2}x^2y^2$ on $L^2(\mathbb{R}^2)$, defined as the Friedrichs operator of its form on $C_0^\infty(\mathbb{R}^2)$.

1. As quadratic forms,
$$H \;\geq\; \tfrac{1}{2}K, \qquad K = -\frac{\hbar^2}{2m}\Delta + \frac{\hbar g}{2\sqrt{m}}\,(|x| + |y|).$$
2. H has compact resolvent: its spectrum is a sequence $E_0 < E_1 < E_2 < \cdots$ of eigenvalues of finite multiplicity tending to infinity.
3. E_0 is simple with a strictly positive eigenfunction, so $\Delta = E_1 - E_0 > 0$, and $H - E_0$ satisfies the spectral-gap part of the Jaffe–Witten conditions with $0 < m = \Delta < \infty$.
4. $E_n(\hbar, m, g) = E_*\, e_n$ with e_n the eigenvalues of $\frac{1}{2}(-\Delta + x^2y^2)$ and
$$E_* = \hbar^{4/3}g^{2/3}m^{-2/3}, \qquad \Delta = \delta_1\,\hbar^{4/3}g^{2/3}m^{-2/3}, \quad \delta_1 = e_1 - e_0 > 0.$$
Hence $\Delta \to 0$ as $\hbar \to 0$ and as $g \to 0$. At $g = 0$ the spectrum is $[0, \infty)$ with no eigenvalue. The classical Hamiltonian $|p|^2/2m + g^2x^2y^2/2$ has range $[0, \infty)$, and the motions $y = p_y = 0$, $x(t) = x_0 + p_x t/m$ escape to infinity along the valley with arbitrarily small energy.

Proof. **Oscillator bound.** For $\phi \in C_0^\infty(\mathbb{R})$ and $\mu, \nu > 0$, with $\alpha = \sqrt{\nu/\mu}$, integration by parts gives $\int |\phi' + \alpha y\phi|^2 dy = \int |\phi'|^2 + \alpha^2 \int y^2|\phi|^2 - \alpha \int |\phi|^2$, so
$$\int (\mu|\phi'|^2 + \nu y^2|\phi|^2)dy \;\geq\; \sqrt{\mu\nu}\int |\phi|^2 dy.$$
With $\mu = \hbar^2/2m$ and $\nu = \frac{1}{2}m\omega^2$ this is the zero-point energy $\hbar\omega/2$ of the oscillator $-\frac{\hbar^2}{2m}\partial_y^2 + \frac{1}{2}m\omega^2y^2$.

Slicing. Fix x and apply the bound to $\phi = \psi(x, \cdot)$ with $\nu = g^2x^2/2$, that is $m\omega_x^2 = g^2x^2$, $\omega_x = g|x|/\sqrt{m}$, $\sqrt{\mu\nu} = \hbar g|x|/(2\sqrt{m})$. Integrating in x (Fubini):
$$\int \Big[\frac{\hbar^2}{2m}|\partial_y\psi|^2 + \frac{g^2}{2}x^2y^2|\psi|^2\Big] \;\geq\; \int \frac{\hbar g|x|}{2\sqrt{m}}|\psi|^2,$$

and symmetrically with $x \leftrightarrow y$. Writing $2q_H(\psi) = \int \frac{\hbar^2}{2m}|\nabla\psi|^2 + \{\text{the } y\text{-slice}\} + \{\text{the } x\text{-slice}\}$ and using the two slice bounds gives $2q_H \geq q_K$ on C_0^∞, which is Simon's display (5) with the constants restored. The inequality extends to the form domain of H by density and shows $Q(H) \subset Q(K)$; the min-max values then satisfy $\mu_n(H) \geq \frac{1}{2}\mu_n(K)$.

Compact resolvent. Let $W \geq \gamma|x|$ be continuous with $\gamma > 0$ and $K = -\frac{\hbar^2}{2m}\Delta + W$. On the unit form-ball $\{q_K(\psi) + \|\psi\|^2 \leq 1\}$ the mass outside the ball of radius R is at most $(\gamma R)^{-1}$ and the H^1 norm on the ball is bounded, so the Rellich–Kondrachov theorem gives total boundedness in L^2; thus $(K+1)^{-1/2}$ and $(K+1)^{-1}$ are compact (this is the case needed of Reed–Simon IV, Theorem XIII.67). Here $W = \frac{\hbar g}{2\sqrt{m}}(|x| + |y|) \geq \frac{\hbar g}{2\sqrt{m}}\sqrt{x^2+y^2}$. Since $\mu_n(H) \geq \frac{1}{2}\mu_n(K) \to \infty$, H has compact resolvent as well (Reed–Simon IV, Theorem XIII.16).

Simplicity. The Feynman–Kac formula $(e^{-tH/\hbar}\psi)(x) = \mathbb{E}^x[\exp(-\hbar^{-1}\int_0^t V(b_s)ds)\psi(b_t)]$ with V continuous and finite along paths shows that $e^{-tH/\hbar}$ is positivity improving for every $t > 0$, and Reed–Simon IV Theorem XIII.44 then gives a simple ground state with strictly positive eigenfunction.

Scaling. The unitary dilation with $\ell^6 = \hbar^2/(mg^2)$ gives $U_\ell^{-1} H U_\ell = E_* \frac{1}{2}(-\Delta + x^2y^2)$ directly, fixing the stated dimensionless operator and all factors of two. At $g = 0$ the Fourier transform diagonalizes $-\hbar^2\Delta/2m$ as multiplication by $\hbar^2|p|^2/2m$. The classical statements are direct: on the invariant set $\{y = p_y = 0\}$ Hamilton's equations reduce to free motion in x. □

Mechanism. The valleys $\{xy = 0\}$ are unbounded and the classical phase-space volume below any positive energy is infinite, so a classical particle escapes at arbitrarily low energy. The transverse zero-point energy $\hbar\omega_x/2 = \hbar g|x|/(2\sqrt{m})$ grows linearly along a valley and confines the quantum particle. That confining term is the product of the two non-commutativities: $\hbar$ from $[x,p] = i\hbar$ and g from the quartic coupling, which is the squared commutator of the gauge potentials in Section 3.2. Removing either factor removes the gap, with the exponents 4/3 and 2/3 of Theorem 5(4).

3.2 The SU(2) matrix model and its abelian valleys

Let $T^a = \sigma^a/2$, so $\operatorname{tr} T^a T^b = \frac{1}{2}\delta^{ab}$ and $[T^a, T^b] = i\epsilon^{abc}T^c$; let $X_i = x_i^a T^a$ with $\vec{x}_i \in \mathbb{R}^3$, $i = 1, \ldots, D$. Then $[X_i, X_j] = i(\vec{x}_i \times \vec{x}_j)^c T^c$ and $-\operatorname{tr}[X_i, X_j]^2 = \frac{1}{2}|\vec{x}_i \times \vec{x}_j|^2$. The Yang–Mills quantum-mechanical Hamiltonian is

$$H = -\frac{\hbar^2}{2m}\Delta_{\mathbb{R}^{3D}} + \frac{g^2}{2}\sum_{i<j}|\vec{x}_i \times \vec{x}_j|^2 = -\frac{\hbar^2}{2m}\Delta_{\mathbb{R}^{3D}} - g^2\sum_{i<j}\operatorname{tr}[X_i, X_j]^2;$$

other trace or sum conventions multiply g^2 by a fixed number and every energy by its cube root.

Theorem 6. For $D \geq 2$:

1. For every ordered pair $i \neq j$, as forms, $-\frac{\hbar^2}{2m}\Delta_{\vec{x}_j} + \frac{g^2}{2}|\vec{x}_i \times \vec{x}_j|^2 \;\geq\; \frac{\hbar g}{\sqrt{m}}|\vec{x}_i|$.
2. As forms on $L^2(\mathbb{R}^{3D})$,

$$H \;\geq\; \tfrac{1}{2}\Big[-\frac{\hbar^2}{2m}\Delta_{\mathbb{R}^{3D}} + \frac{\hbar g}{\sqrt{m}}\sum_{i=1}^{D}|\vec{x}_i|\Big].$$

3. H has compact resolvent and a simple strictly positive ground state; its gap is $\Delta = \delta_1^{(D)}\,\hbar^{4/3}g^{2/3}m^{-2/3} > 0$, with the same $\hbar \to 0$ and $g \to 0$ conclusions as Theorem 5.
4. The $SO(3)$ action $(U(R)\psi)(\vec{x}_1, \ldots) = \psi(R^{-1}\vec{x}_1, \ldots)$ commutes with H; the invariant subspace, which Gauss's law selects in the constant-mode truncation of the gauge theory, reduces H, contains the ground state, and has gap $\Delta_{\text{inv}} \geq \Delta > 0$.
5. The potential vanishes exactly on the commuting configurations $\{\vec{x}_i = \lambda_i \hat{n}\}$, where all X_i lie in one Cartan subalgebra. For $D = 1$ there is no pair, $V \equiv 0$, and $H = -\hbar^2\Delta_{\mathbb{R}^3}/2m$ has spectrum $[0, \infty)$.

Proof. (1) For fixed $\vec{x}_i \neq 0$ choose an orthonormal frame of $\mathbb{R}^3_{\vec{x}_j}$ with third axis along $\vec{x}_i$. Then $|\vec{x}_i \times \vec{x}_j|^2 = |\vec{x}_i|^2(u_1^2 + u_2^2)$; drop the kinetic term in u_3 and apply the oscillator bound in u_1 and in u_2 with $\nu = g^2|\vec{x}_i|^2/2$, each giving $\hbar g|\vec{x}_i|/(2\sqrt{m})$; the complement $\vec{x}_i = 0$ is a null set. Integrate over the remaining variables. (2) Sum the pair inequalities over all $D(D-1)$ ordered pairs with weight $w = 1/(2(D-1))$: each kinetic term T_j appears with total weight $w(D-1) = \frac{1}{2}$ and each unordered pair potential with weight $2w = 1/(D-1) \leq 1$. Then $H = \sum_{i\neq j} w(T_j + V_{ij}) + \frac{1}{2}\sum_j T_j + (1 - \frac{1}{D-1})\sum_{i<j} V_{ij}$, where both remainders are nonnegative forms; dropping the potential remainder and using (1) gives (2). (3) The bracket in (2) has $W = \frac{\hbar g}{\sqrt{m}}\sum_i |\vec{x}_i| \geq \frac{\hbar g}{\sqrt{m}}|x|_{\mathbb{R}^{3D}}$, so the compact-resolvent, simplicity and scaling steps of Theorem 5 apply verbatim with $n = 3D$, $k = 4$. (4) $U(R)$ is unitary, preserves C_0^∞, commutes with each $\Delta_{\vec{x}_i}$ and preserves $|\vec{x}_i \times \vec{x}_j|$; the Haar average $\int U(R)\, dR$ is the orthogonal projection onto the invariant subspace and commutes with $(H+1)^{-1}$, so the restriction is self-adjoint with compact resolvent. The strictly positive simple ground state ψ_0 satisfies $U(R)\psi_0 = c(R)\psi_0$ with $c(R) > 0$ and $|c(R)| = 1$, hence is invariant, so $E_0^{\mathrm{inv}} = E_0$ and $E_1^{\mathrm{inv}} \geq E_1$. (5) $\vec{x}_i \times \vec{x}_j = 0$ for all pairs iff all nonzero vectors are parallel. □

The abelian and the two-dimensional cases. Along the commuting configurations the matrices commute and the potential is flat; writing $\vec{x}_i = \lambda_i \hat{n} + \vec{\xi}_i$ with $\vec{\xi}_i \perp \hat{n}$, the transverse curvature of the potential grows quadratically in $|\lambda|$, which is what (1) exploits. An abelian gauge group has *only* commuting configurations: its matrix model has $V \equiv 0$ and the free spectrum $[0, \infty)$. This is the gapless *free abelian matrix model*, not a theorem about every commutative algebra or every abelian gauge theory. The free-field comparison and compact abelian exceptions are treated in §5. The temporal-gauge constant modes of a gauge theory in d spacetime dimensions are $D = d - 1$ spatial components, so the zero-point mechanism needs $d \geq 3$; for $d = 2$ the single matrix has no partner, in agreement with §5, where two-dimensional Yang–Mills has no local excitations at all and its discrete flux spectrum comes from the compact holonomy, a different mechanism.

3.3 Conditional orbit-area bounds and state counting

For the same classical scalar Hamiltonian, freeze $x \neq 0$ and let $E_y = p_y^2/(2m) + m\omega_x^2 y^2/2$, $\omega_x = g|x|/\sqrt{m}$. The transverse oscillator ellipse has area

$$J_y = \oint p_y\, dy = \frac{2\pi E_y}{\omega_x} = \frac{2\pi\sqrt{m}E_y}{g|x|}.$$

Indeed its semiaxes are $\sqrt{2mE_y}$ and $\sqrt{2E_y/(m\omega_x^2)}$. The frozen-oscillator identity is exact; adiabatic conservation along a valley is only a leading-order reading when $|\dot{x}/x| \ll \omega_x$, or $|x| \gg x_{\mathrm{ad}}(E) = (2E)^{1/4}g^{-1/2}$. The original Landau–Lifshitz §49 comparison was metadata only; the bound below requires no adiabatic approximation.

Conditional sublevel bound (former G08 Theorem 2). Supply an arbitrary action coefficient $a_0 > 0$ and name the restriction $\mathsf{F}(a_0)$ on instantaneous admitted states by $J_y \geq \pi a_0$ when $x \neq 0$ and $J_x \geq \pi a_0$ when $y \neq 0$. Then

$$E_y \geq \frac{a_0 g|x|}{2\sqrt{m}}, \qquad |x|, |y| \leq \frac{2\sqrt{m}E}{a_0 g}.$$

This follows directly from the ellipse formula and $E_y \leq E$; interchange x, y for the other bound. Momenta also satisfy $|p| \leq \sqrt{2mE}$, so the restricted sublevel set is bounded. Without the restriction, $y = p_y = 0$, $x(t) = x_0 + p_x t/m$ escapes at any positive kinetic energy. At $a_0 = \hbar$, the numerical lower bound is the oscillator inequality in Theorem 5. Its quantum proof, however, uses a Hilbert space, derivatives, normalization and the canonical momentum operator, all additional structure.

The classical restriction alone has no discrete spectrum and need not be invariant under the full coupled dynamics. It does not even force a positive infimum of the instantaneous admitted energies: at $x = y = r > 0$ choose $p_x^2 = p_y^2 = a_0 g\sqrt{m}\, r$. Both area bounds hold while $H = a_0 g r/\sqrt{m} + g^2 r^4/2 \to 0$ as $r \downarrow 0$. Thus the former claim that a classical area floor itself proves a spectral gap is withdrawn. A dynamical, invariant trajectory class would require a separate argument.

For the following count set $a_0 = \hbar$ solely to compare with Theorem 5, and define $E_* = \hbar^{4/3} g^{2/3} m^{-2/3}$ and $X(E) = 2\sqrt{m}E/(\hbar g)$. This comparison does not derive physical $\hbar$.

Floor-admitted phase-space volume. Without the floor the phase-space volume $|\{H_{\mathrm{cl}} \le E\}|$ is infinite for every $E > 0$ (Simon, Ann. Phys. 146 (1983) 209–220, p. 210, passage read). With the floor it is finite, and it can be computed with both valleys included. Write $R = \sqrt{2E}/g$, so that the potential is below E exactly where $|xy| \le R$; the momenta at (x, y) fill a disc of area $2\pi m(E - \frac{g^2}{2}x^2y^2)$. By the conditional sublevel bound every admitted state lies in the box $|x|, |y| \le X(E)$. For $X(E)^2 \ge R$, that is $E \ge E_*/2$, the box carries the volume

$$P(E) = \iint_{|x|,|y|\le X} 2\pi m\Big(E - \frac{g^2}{2}x^2y^2\Big)_+ dx\, dy = 8\pi m\, ER\Big[\frac{8}{9} + \frac{2}{3}\ln\frac{X^2}{R}\Big]:$$

in each quadrant the strip $0 \le x \le R/X$, where the y-integral of $E - \frac{g^2}{2}x^2y^2$ runs to X, contributes $\frac{8}{9}ER$, and the region $R/X \le x \le X$, where it runs to R/x and equals $\frac{2}{3}ER/x$, contributes $\frac{2}{3}ER\ln(X^2/R)$. Inside the box the floor removes, at each (x, p_x) with $x \ne 0$, the ellipse $E_y < \hbar\omega_x/2$ of area $\pi\hbar$ in the (y, p_y) plane (the area identity above), and at each (y, p_y) with $y \ne 0$ the ellipse $E_x < \hbar\omega_y/2$ of the same area. The admitted volume $V(E) = |\{H_{\mathrm{cl}} \le E\} \cap \mathsf{F}(\hbar)|$ therefore obeys $0 \le P - V \le 2 \cdot \pi\hbar \cdot 2X \cdot 2\sqrt{2mE} = 16\sqrt{2}\,\pi m E^{3/2}/g$. With $ER = \sqrt{2}E^{3/2}/g$ and $X^2/R = 2\sqrt{2}\,(E/E_*)^{3/2}$,

$$V(E) = \frac{8\sqrt{2}\,\pi m}{g}\,E^{3/2}\Big[\ln\frac{E}{E_*} + \theta(E)\Big], \qquad \ln 2 - \frac{10}{9} \le \theta(E) \le \ln 2 + \frac{8}{9}.$$

Since $\sqrt{R} = x_{\mathrm{ad}}(E)$, the hyperbola $|xy| = R$ crosses the diagonal at the adiabatic scale defined above. To leading order V is twice the one-valley integral of $(8\pi/3)\sqrt{2}\,mE^{3/2}/(g|x|)$, the transverse ellipse area $2\pi(E - p_x^2/2m)/\omega_x$ integrated over $|p_x| \le \sqrt{2mE}$, over $x_{\mathrm{ad}}(E) \le |x| \le X(E)$. A floor κh_P with any $\kappa > 0$ in place of $h_P/2$ replaces X by $X/(2\kappa)$, leaves the bound on $P - V$ unchanged (the excluded ellipses grow by 2κ as the box shrinks by $1/(2\kappa)$) and shifts both bounds on θ by $-\frac{4}{3}\ln(2\kappa)$, so the coefficient of $E^{3/2}\ln E$ is independent of the floor value.

Comparison with the exact count. For the operator $-\Delta + |xy|^\alpha$, Simon proves the limit $E^{-1-1/\alpha}(\ln E)^{-1}N(E) \to \pi^{-1}$ (J. Funct. Anal. 53 (1983) 84–98, Theorem 1.4 and the display after Theorem 1.5, pp. 85–86, passage read). At $\alpha = 2$ this is $N(E) \simeq \pi^{-1}E^{3/2}\ln E$, the growth also stated in Ann. Phys. 146, p. 212, end of Section 2. The dilation $x \mapsto \ell x$ with $\ell^6 = \hbar^2/(mg^2)$ carries $H = -\frac{\hbar^2}{2m}\Delta + \frac{g^2}{2}x^2y^2$ to $\frac{E_*}{2}(-\Delta + x^2y^2)$, because both coefficients become $\hbar^2/(2m\ell^2) = g^2\ell^4/2 = E_*/2$. Hence $N_H(E) \simeq (2\sqrt{2}/\pi)(E/E_*)^{3/2}\ln(E/E_*)$, while the floor count gives $V(E)/h_P^2 = (2\sqrt{2}/\pi)(E/E_*)^{3/2}[\ln(E/E_*) + \theta(E)]$. The leading terms agree, constant included. The identification $N \approx V/h_P^2$ is the Weyl heuristic, and the agreement is a comparison of two computed leading terms; Simon's proof runs through a Feynman–Kac lower bound and sliced-bread upper bounds. The leading coefficient is independent of the chosen positive floor multiple. Agreement therefore cannot select its value or establish a classical area restriction as the physical origin of the quantum count. Simon (p. 86) calls a general geometric interpretation of the constant π^{-1} an open question; no search was run for later literature that supplies one, and no novelty is claimed.

Semiclassical channel levels. With $J_y = (n + \frac{1}{2})h_P$ conserved, the valley motion has effective Hamiltonian $p_x^2/2m + \kappa_n|x|$, $\kappa_n = (n + \frac{1}{2})\hbar g/\sqrt{m}$, and Bohr–Sommerfeld quantization

$\oint p_x\,dx = (N+\frac{1}{2})h_P$ with $\oint p_x\,dx = \frac{8\sqrt{2m}}{3}E^{3/2}/\kappa_n$ gives

$$E^{\rm BS}_{n,N} = E_*\Big[\frac{3\pi(2n+1)(2N+1)}{16\sqrt{2}}\Big]^{2/3}, \qquad E_* = \hbar^{4/3}g^{2/3}m^{-2/3}.$$

The unit E_* is the exact one of Theorem 5(4); the pure numbers are semiclassical estimates of e_n and are labelled as such. The estimate describes one channel, the x-axis with its two half-valleys. The y-axis gives the same levels, so counting both channels doubles every level, and the coupling of the channels and of different n near the origin, where the adiabatic reading fails, is ignored. The quantization rules add a discrete set of positive approximate channel levels. They are extra assumptions; they are not a spectral consequence of the classical area restriction. The splitting of the doublets and the simplicity of the ground state, which the Jaffe–Witten conditions need, come from Theorem 5(3). As a consistency check, each channel and each n contribute $\frac{4\sqrt{2}}{3\pi}(E/E_*)^{3/2}/(n+\frac{1}{2})$ levels below E; summed over both channels and over n up to the adiabatic limit $n_* \simeq 2^{-1/4}(E/E_*)^{3/4}$, where the turning point E/κ_n reaches $x_{\rm ad}(E)$, they give $(2\sqrt{2}/\pi)(E/E_*)^{3/2}\ln(E/E_*)$ to leading order, the same leading term as the volume count.

For $SU(2)$, each of the two frozen transverse oscillator directions can instead be assigned area at least πa_0. Their sum gives $E_\perp \geq a_0 g|\vec{x}_i|/\sqrt{m}$ and the corresponding classical sublevel bound $|\vec{x}_i| \leq \sqrt{m}E/(a_0 g)$. The quantum counterpart is Theorem 6(1); imposing a classical restriction does not reproduce its operator domain or state space. For an abelian matrix model, or $D = 1$, there is no transverse restoring oscillator: this restriction supplies no valley confinement, and the quantum model is free.

3.4 Torus truncation and the uniformity problem

Proposition 7. For SU(2) Yang–Mills on a spatial torus of side L with action $S = -\frac{\hbar}{4g^2}\int F^a_{\mu\nu}F^{a\mu\nu}\,d^3x\,c\,dt$ (g dimensionless, $[A] = \text{length}^{-1}$), temporal gauge and the constant-mode ansatz $A^a_i(x,t) = a^a_i(t)$, the Hamiltonian is

$$H_0 = \frac{g^2 c}{2\hbar L^3}\sum_{i=1}^{3}|\vec{p}_i|^2 + \frac{\hbar c L^3}{2g^2}\sum_{i<j}|\vec{a}_i\times\vec{a}_j|^2, \qquad [a^a_i, p^b_j] = i\hbar\delta_{ij}\delta^{ab},$$

which is Theorem 6 with $D = 3$, $m = \hbar L^3/(g^2 c)$ and $g_B^2 = \hbar c L^3/g^2$. Its energy unit is

$$E_* = \hbar^{4/3}g_B^{2/3}m^{-2/3} = \frac{g^{2/3}\,\hbar c}{L},$$

so the unrestricted zero-mode gap is $\delta_1^{(3)}g^{2/3}\hbar c/L$. On the Gauss-invariant sector write instead $\Delta_{0,\rm inv}(L) = \delta^{(3)}_{\rm inv}g^{2/3}\hbar c/L$, where $\delta^{(3)}_{\rm inv} \geq \delta^{(3)}_1 > 0$; equality is not asserted. Both truncation gaps decay as $1/L$ at fixed coupling.

Proof. With $F^a_{0i} = c^{-1}\dot{a}^a_i$ and $F^a_{ij} = (\vec{a}_i\times\vec{a}_j)^a$ the Lagrangian is $\frac{\hbar L^3}{2g^2 c}\sum|\dot{\vec{a}}_i|^2 - \frac{\hbar c L^3}{4g^2}\sum_{i,j}|\vec{a}_i\times\vec{a}_j|^2$; the Legendre transform and $\frac{1}{4}\sum_{i,j} = \frac{1}{2}\sum_{i<j}$ give H_0, and the exponent bookkeeping is $\hbar^{4/3+1/3-2/3}c^{1/3+2/3}L^{1-2}g^{-2/3+4/3}$. □

Lüscher's small-volume expansion (Nucl. Phys. B219 (1983) 233; abstract) states that the torus spectrum is discrete and expands in powers of $g^{2/3}$ with g the renormalized coupling; Proposition 7 is the leading constant-mode model used in that approximation, not a uniform error theorem for the full field Hamiltonian, and Lüscher–Münster (Nucl. Phys. B232 (1984) 445; abstract) carry it to one loop for SU(2) and locate the crossover to large-volume behaviour near $z = M(0^+)L \simeq 2$. Jaffe and Witten write (p. 7 of the local PDF) that "no present ideas point the direction to establish the existence of a mass gap that is uniform in the volume". Proposition 7 shows the difficulty in the solved sector: $\Delta_0(L) > 0$ for every L with $\inf_L \Delta_0(L) = 0$. Uniformity would need $g(L)^{2/3}/L$ bounded below, outside the weak-coupling regime where the truncation is

justified; the compensation must come from the crossover regime, where the expansion no longer applies. In the language of the dimensional theorem cited in §1, L is a fixed apparatus length that enters the monomial and makes the finite-volume floor non-universal, and the uniformity requirement is exactly the demand that the floor be independent of it.

The supersymmetric contrast identifies the mechanism once more: de Wit, Lüscher and Nicolai (Nucl. Phys. B320 (1989) 135; abstract) prove that the supersymmetric matrix-model Hamiltonians have continuous spectrum starting at zero, through low-energy states along the same abelian valleys. The fermionic zero-point energy cancels the bosonic $\hbar\omega/2$ per transverse mode, so the confining term of Theorem 5 vanishes. The uncancelled bosonic transverse energy is the confinement mechanism in Theorems 5–6. Its cancellation removes this argument in the cited supersymmetric models.

The Feshbach treatment states the separate complement, Schur and effective-gap estimates needed to transport this truncation into a renormalized field theorem. The three-dimensional scaling problem uses $D = 2$, with its own physical construction and order of limits.

4. Gap-to-action units and the field dimension

Gap-to-action unit (C132a; former G08 Proposition 4). Let the classical model with fixed constants m, g (dimension vectors $d_m = (1, 0, 0)$, $d_g = (\frac{1}{2}, -1, -1)$) be enlarged by any fixed constants, and suppose the enlarged theory has a positive finite gap Δ depending on the fixed constants only. Then the theory has the action unit

$$A = \Delta^{3/4}\, m^{1/2}\, g^{-1/2},$$

and conversely an action unit A gives the energy unit $A^{4/3} g^{2/3} m^{-2/3}$.

Proof. Energy has dimension vector $(1, 2, -2)$ and action $(1, 2, -1)$. The system $a\, d_m + b\, d_g + c\, (1, 2, -2) = (1, 2, -1)$ has the unique solution $a = \frac{1}{2}$, $b = -\frac{1}{2}$, $c = \frac{3}{4}$: the length and time components give $-b + 2c = 2$, $-b - 2c = -1$, hence $c = \frac{3}{4}$, $b = -\frac{1}{2}$, and the mass component gives $a = \frac{1}{2}$. So $m^{1/2} g^{-1/2} \Delta^{3/4}$ has action dimension whatever the extra constants are. The converse energy monomial follows from Theorem 3 with $k = 4$, but dimensional permission does not establish any spectrum or positive gap. □

A supplied positive finite gap gives this action unit, and a supplied action unit gives an energy unit. These are equivalent *unit constructions*. Theorem 5 supplies the additional spectral content $\Delta = \delta_1 E_*$, $0 < \delta_1 < \infty$, for the quantized model.

Field dimension (C132b; Proposition 10 and former G08 Proposition 5). Write the Euclidean Yang–Mills action with the gauge field as a connection, $D = \partial + A$, $[A] = \text{length}^{-1}$, $x_0 = ct$:

$$S_E = \frac{1}{4 g_{\text{cl}}^2} \int F^a_{\mu\nu} F^a_{\mu\nu}\, d^d x, \qquad [1/g_{\text{cl}}^2] = \text{action} \cdot \text{length}^{4-d}.$$

The weight $e^{-S_E/\hbar}$ depends on the coupling through $\lambda_d = \hbar g_{\text{cl}}^2$, of dimension length^{d-4}; in $d = 4$ the repository writes $g^2 = \hbar g_{\text{cl}}^2$, a pure number (the trace-normalized action has the same dimension). With the speed c and a gap Δ:

1. In $d = 2$ and $d = 3$ the pair $(1/g_{\text{cl}}^2, c)$ contains neither an action unit nor an energy unit. Adjoining Δ gives exactly one action unit, adjoining $\hbar$ gives exactly one energy unit, and the same gap/action unit construction holds, without a positivity theorem.
2. In $d = 4$, $1/g_{\text{cl}}^2$ is itself an action unit. The constants $(1/g_{\text{cl}}^2, c, \Delta)$ are dimensionally independent and their only action monomial is $1/g_{\text{cl}}^2$, with Δ at exponent zero. The constants $(1/g_{\text{cl}}^2, c, \hbar)$ contain no energy, length or time unit. So "gap ⇒ action unit" holds trivially and "action unit ⇒ gap" fails.

d	classical constants (with c for Yang–Mills)	action unit with Δ adjoined	energy unit with $\hbar$ adjoined
2	$1/g_{\rm cl}^2 \sim (1,4,-1)$	$(1/g_{\rm cl}^2)^{1/3} c^{-2/3} \Delta^{2/3}$	$\hbar^{3/2} g_{\rm cl}\, c = \hbar c\, \lambda_2^{1/2}$
3	$1/g_{\rm cl}^2 \sim (1,3,-1)$	$(1/g_{\rm cl}^2)^{1/2} c^{-1/2} \Delta^{1/2}$	$\hbar^2 g_{\rm cl}^2\, c = \hbar c\, \lambda_3$
4	$1/g_{\rm cl}^2 \sim (1,2,-1)$, an action	$1/g_{\rm cl}^2$ itself; Δ enters with exponent 0	none: $\hbar g_{\rm cl}^2 = g^2$ is a pure number
mechanics, $k=-2$	m, $\lambda \sim (1,4,-2)$	$\sqrt{m\lambda}$ already classical; Δ adds nothing	none: $2m\lambda/\hbar^2$ is a pure number

Proof. Dimension vectors are over (mass, length, time); action is $(1,2,-1)$, energy $(1,2,-2)$, $c \sim (0,1,-1)$. Solve $a\, d_{1/g_{\rm cl}^2} + b\, d_c + c'\,(1,2,-2) = (1,2,-1)$. For $d=3$, adding the length and time equations gives $2a = 1$, then $c' = \frac{1}{2}$, $b = -\frac{1}{2}$; for $d=2$ the sum gives $3a = 1$, then $c' = \frac{2}{3}$, $b = -\frac{2}{3}$; for $d=4$ it gives $a = 1$, then the mass equation gives $c' = 0$ and the length equation $b = 0$. The determinants of the three dimension vectors are -3, -2, -1 for $d = 2,3,4$, so each solution is unique. With $\hbar \sim (1,2,-1)$ in place of Δ the determinants are -2 and -1 for $d = 2,3$, and the displayed energy monomials have vector $(1,2,-2)$ by direct addition. In $d = 2,3$ the pair $(1/g_{\rm cl}^2, c)$ alone fails: the mass and length equations fix $a = 1$ and $b = -2$ (respectively $b = -1$), and the time component is then 1 (respectively 0), never -1 or -2. In $d = 4$, $1/g_{\rm cl}^2$ and $\hbar$ have the same vector, so the span is that of $(1,2,-1)$ and $(0,1,-1)$; $a(1,2,-1) + b(0,1,-1) = (1,2,-2)$ forces $a = 1$ (mass), $b = 0$ (length) and then $-1 = -2$ (time), and the targets $(0,1,0)$ and $(0,0,1)$ fail at the time component in the same way. The $k=-2$ row is §6 below. □

In $d = 2,3$ the quantum theory in infinite volume has the constants $(\hbar, c, \lambda_d)$, which are independent, so a gap that depends on them alone is $\Delta = \kappa\, \hbar c\, \lambda_d^{1/(4-d)}$ with a pure number κ (the dimensional theorem cited in §1). The action unit of the table is then $\kappa^{2/3}\hbar$ in $d=2$ and $\kappa^{1/2}\hbar$ in $d=3$: the gap returns $\hbar$ up to a pure number.

Two facts fix the reading of the table. First, the overall factor $1/g_{\rm cl}^2$ drops out of the classical field equations $D_\mu F_{\mu\nu} = 0$ in every dimension (the field normalization above), and the dilation $A \mapsto sA(sx)$ maps solutions to solutions in every d while multiplying the action by s^{4-d}. The classical field equations are therefore dilation covariant in every dimension; they supply no chosen solution length. The normalization $1/g_{\rm cl}^2$ still converts solution geometry into action and energy. The difference between the rows becomes physical through the quantum weight $e^{-S_E/\hbar}$, whose coefficient $1/\lambda_d = 1/(\hbar g_{\rm cl}^2)$ has dimension length^{4-d}. In $d = 2,3$ the weight carries a length and, with c, an energy, and a gap is a pure number times that energy. In $d=4$ the weight carries the pure number $1/g^2$ and no length. Second, $d=4$ is action-critical in the sense of §6, and shares its row type with the inverse-square potential: the classical constants own an action unit and no energy unit. This is the dimensional reason the four-dimensional problem is one-directional. A gap needs a length that the constants of the weight do not contain; dimensional transmutation generates it from the regulator (the field normalization above), and the unique action monomial formed using the classical coupling, c and that gap still equals the $1/g_{\rm cl}^2 = \hbar/g^2$ already present. Along the continuum trajectory of asymptotic freedom $g^2(a) \to 0$, so this bare action unit grows without bound in units of $\hbar$; the continuum theory keeps $\hbar$, c and the transmuted scale, the dimensionless coupling having been traded for that scale. The same normalization distinction applies to C132a: the trajectories of $H_{\rm cl}$ depend on g^2/m alone and m normalizes the action, while an actual spectral gap additionally depends on the specified quantization; the unit conversion itself does not.

5. Free fields, compact groups and the exact circle theory

Let H be the second quantization on Fock space of a one-particle operator $\hbar\Omega$ with $\Omega \geq 0$ the classical frequency operator. Then $\operatorname{spec}(H) \setminus \{0\}$ has infimum $\hbar \inf(\operatorname{spec}\Omega \setminus \{0\})$. In particular a free quantum field has a gap if and only if its classical dispersion relation has one.

Proof. On the n-particle sector H acts as $\hbar \sum_{j \leq n} \Omega_j$; its spectrum is the closure of n-fold sums of spectral values of $\hbar\Omega$, and the one-particle sector attains the infimum. □

Corollary 9 (Maxwell). In a periodic box of side L, Coulomb gauge and two transverse polarizations give $H = \sum_{k \neq 0, \sigma} \hbar c|k|\, n_{k\sigma}$ after normal ordering, $k \in (2\pi/L)\mathbb{Z}^3$: the gap is $2\pi\hbar c/L$ and tends to zero as $L \to \infty$, where the spectrum is $[0, \infty)$. A scale-covariant classical dispersion $\omega(sk) = s\omega(k)$ has infimum zero in infinite volume; a mass term gives $\omega = \sqrt{c^2k^2 + \mu^2c^4/\hbar^2}$ and the supplied gap μc^2, and gauge invariance excludes the Maxwell mass term $A_\mu A^\mu$. The constant modes are free ($k = 0$, spectrum $[0, \infty)$ already at finite L) in the non-compact theory; in compact $U(1)$ they live on a torus and contribute energies proportional to $\hbar c/L$ as well. Free quantization therefore generates no scale: this free compact finite-volume gap is $\hbar c/L$ times a number.

Compact abelian theories. Compactness admits topological sectors absent from non-compact free Maxwell theory. Osterwalder–Seiler's compact-group strong-coupling construction, Polyakov's three-dimensional monopole mechanism, Göpfert–Mack's all-coupling Villain gap, and the four-dimensional weak-coupling Coulomb phase have different operators and limiting assumptions. Read levels and precise references remain in B78. The Villain treatment retains the normalized measure bounds and distinguishes the fixed-physical-coupling massless limit from Göpfert–Mack's massive Gaussian scaling. A gapped lattice theory can therefore have a massless continuum limit. The former Debye formula with a dimensionless β times a^{-3} was not dimensionally a mass-squared formula and is withdrawn in this normalization; use the canonical trajectory and correlation bounds instead of that unchecked prefactor.

The exact two-dimensional holonomy spectrum

Proposition 11. Pure Yang–Mills on a spatial circle of circumference L, in temporal gauge with Gauss's law imposed, has as its only gauge-invariant degree of freedom the holonomy $U \in G$ up to conjugation. The physical Hilbert space is $L^2(G)^G$, the class functions, and with $S = \frac{1}{2g_{\mathrm{cl}}^2} \int \operatorname{tr} F^2$ the Hamiltonian is $H = \frac{g_{\mathrm{cl}}^2 \hbar^2 c L}{2} C$, where C is the quadratic Casimir acting on class functions. By Peter–Weyl the eigenfunctions are the characters χ_R and

$$E_R = \frac{g_{\mathrm{cl}}^2 \hbar^2 c L}{2} C_2(R), \qquad E_{\mathrm{trivial}} = 0, \qquad \mathrm{gap} = \frac{g_{\mathrm{cl}}^2 \hbar^2 c L}{2} C_2(R_{\min}) > 0;$$

for $SU(N)$ with $\operatorname{tr} T^a T^b = \frac{1}{2}\delta^{ab}$, $C_2(\mathrm{fund}) = (N^2 - 1)/(2N)$, and for $U(1)$, $E_n = \frac{e^2\hbar^2 c L}{2} n^2$ with $n \in \mathbb{Z}$ from compactness.

Proof. Gauss's law $D_x E = 0$ makes the electric field covariantly constant along the circle, so on class functions of the holonomy E acts as the left-invariant derivative and $\int_0^L \operatorname{tr} E^2\, dx$ as L times the Laplacian on the group; $H = \frac{g_{\mathrm{cl}}^2}{2} \int \operatorname{tr} E^2$ follows from $E = \dot{A}/g_{\mathrm{cl}}^2$, and restoring $\hbar, c$ uses $[g_{\mathrm{cl}}^2 \hbar] = L^{-2}$ from §4. Characters diagonalize the Laplacian on class functions with eigenvalue the Casimir. □

The states are electric flux lines wrapping the circle, E_R/L is a string tension, and there are no propagating excitations: the theory has no local degrees of freedom. The gap grows linearly with L and diverges as $L \to \infty$, so the Jaffe–Witten finiteness requirement $m < \infty$ fails in infinite volume, where the finite-energy Hilbert space is the vacuum alone. Abelian and non-abelian groups behave identically here; the commutator plays no role in $d = 2$ because there are no transverse spatial directions, in agreement with $D = 1$ in Theorem 6. Exact solvability on the Euclidean side is Migdal's and Witten's (Commun. Math. Phys. 141 (1991) 153); rigorous

constructions of the two-dimensional Yang–Mills measure are Driver's, Gross–King–Sengupta's and Lévy's (B78).

6. Singular quantum mechanics: a scale is not a completion theorem

Proposition 12 (two-dimensional delta potential). For $H = -\frac{\hbar^2}{2m}\Delta - \lambda\delta^2(\vec{r})$ on $\mathbb{R}^2$, $m\lambda/\hbar^2$ is dimensionless and the classical problem has no energy scale. With a momentum cutoff $\hbar\Lambda$ the bound-state condition is

$$1 = \frac{m\lambda}{2\pi\hbar^2}\ln\Big(1 + \frac{\hbar^2\Lambda^2}{2m|E_B|}\Big), \qquad |E_B| = \frac{\hbar^2\Lambda^2}{2m}\Big(e^{2\pi\hbar^2/(m\lambda)} - 1\Big)^{-1}.$$

Removing the cutoff with $\lambda(\Lambda)$ chosen to keep E_B fixed defines the theory by the scale E_B alone. The spectrum is $\{E_B\} \cup [0, \infty)$.

Proof. In momentum space $\tilde{\psi}(p) = \lambda\psi(0)/(p^2/2m + |E_B|)$, and $\psi(0) = \int_{|p|<\hbar\Lambda} \tilde{\psi}\, d^2p/(2\pi\hbar)^2$ gives $1 = \frac{\lambda}{(2\pi\hbar)^2} \cdot 2\pi \int_0^{\hbar\Lambda} \frac{p\,dp}{p^2/2m+|E_B|} = \frac{m\lambda}{2\pi\hbar^2}\ln(1 + \hbar^2\Lambda^2/2m|E_B|)$. □

This chosen nontrivial point interaction has an isolated bound state and excitation threshold $|E_B|$ after shifting the ground energy to zero. Its continuum is the escaping one-particle continuum, not a reconstructed field spectrum. Specifying a nontrivial renormalized interaction supplies E_B; well-defined quantum dynamics also permits the scale-free free operator. Thus this example shows transmutation *within a selected interaction class*, not that every consistent completion must introduce a scale.

Proposition 13 (inverse-square domain and stability distinction). In the s wave of $-\hbar^2\Delta/(2m) - \lambda/r^2$, put $\gamma = 2m\lambda/\hbar^2 > 0$. The reduced expression is $-(\hbar^2/(2m))(u'' + \gamma u/r^2)$ on $L^2(0, \infty)$. For $0 < \gamma \le 1/4$, Hardy's inequality gives a nonnegative Friedrichs extension with spectrum $[0, \infty)$ and no bound state. This names the extension: non-essential self-adjointness does not by itself force a new scale or the same spectrum for every boundary condition.

For $\gamma > 1/4$, write $\nu = \sqrt{\gamma - 1/4}$. The local indices are $1/2 \pm i\nu$, so a self-adjoint origin condition fixes their relative phase, equivalently a length modulo discrete rescaling. The negative eigenvalues form a two-sided geometric tower

$$E_n = -{E_*}^{\text{ext}} \exp(-2\pi n/\nu), \qquad n \in \mathbb{Z}, \qquad {E_*}^{\text{ext}} > 0.$$

They accumulate at zero as $n \to +\infty$ and are unbounded below as $n \to -\infty$. There is no ground state or stable positive excitation gap in this unregularized supercritical model. Keeping only the tower toward zero would silently discard half the extension's spectrum. A short-distance regularization that stabilizes it is an additional model.

The indices follow from $u = r^s$, $s(s-1) + \gamma = 0$. The domain and full tower statements are checked in Dereziński–Richard, §§2.2–2.3 and Theorem 5.5 (passages re-read during consolidation), already used for the singular-domain comparison. In the subcritical range scale-preserving self-adjoint choices exist; at supercritical coupling self-adjointness does not restore semiboundedness. The earlier inference "well-defined dynamics forces a stable scale" is therefore withdrawn.

Classically $m\lambda$ has action-squared units and fall to the centre occurs below angular momentum $\sqrt{2m\lambda}$. This action-critical dimensional comparison with relativistic Coulomb thresholds survives; its domains, quantum dynamics and admissibility premises are different. The remaining Thorn, Camblong, Case and Essin–Griffiths leads retain B78's metadata-level qualification and are not substituted for the domain theorem just read.

7. Consequence for STATE

The matrix gap is proved **after** quantum state/operator structure and physical $\hbar$ are supplied. The classical orbit-area restriction gives a bounded sublevel region and the displayed leading volume count; its Weyl comparison and Bohr–Sommerfeld rules do not turn it into a classical derivation of quantum spectral discreteness. The continuum problem adds control of the

nonzero modes, volume/cutoff limits and physical spectral weight. The exact circle spectrum has global electric flux and no propagating local excitations. These different mechanisms cannot be combined into a universal implication "positive action implies mass gap".

For comparison, constant-force motions at fixed m, F admit $x \mapsto sx$, $t \mapsto s^{1/2}t$, $v \mapsto s^{1/2}v$. Their Galileo comparison $F^2\tau^3/(2m)$ scales by $s^{3/2}$ and area $vF\tau^3/(6m)$ by s^2, giving zero infima when all such alternatives are admitted. A supplied length gives $\sqrt{mF\ell_0^3}$; a supplied speed gives m^2c^3/F; fixed charge and speed give k_e/c. Units do not by themselves bound these observables. The action-selection treatment and Newton insertion proof own those classical arguments.

The former "similarity anomaly" route remains a question: can an independently specified physical composition law exclude every zero-action completion? Associativity, positivity and definability alone have supplied no such theorem; scale-free formulation does not by itself establish similarity covariance. Classical chords and zero/nonzero Gaussian families already compose. The singular examples in §6 illustrate boundary choices and possible instability, not a derivation of physical $\hbar$ from consistency.

Sources for the matrix theorem are Simon (Ann. Phys. **146** (1983) 209–220, pp. 210–212, passage) and his counting theorem (J. Funct. Anal. **53** (1983) 84–98, pp. 85–86, passage). The finite-volume Lüscher/Lüscher–Münster and supersymmetric de Wit–Lüscher–Nicolai comparisons were abstract-level reads; Reed–Simon theorem numbers were corroborated through citing passages rather than a full book audit. B78 and B79 preserve those source boundaries. No new research, independent review or priority claim follows from this consolidation.